

Written homework. Each problem worth 10 points.
Homework Set 1. Total 10 problems

- Homework: INTRO1

- “act rationally”

- What does the word “rationally” means?

Hint: In economics the theory assumes
players/humans are selfish, but act rationally.

Give 1 or 2 other concrete examples.

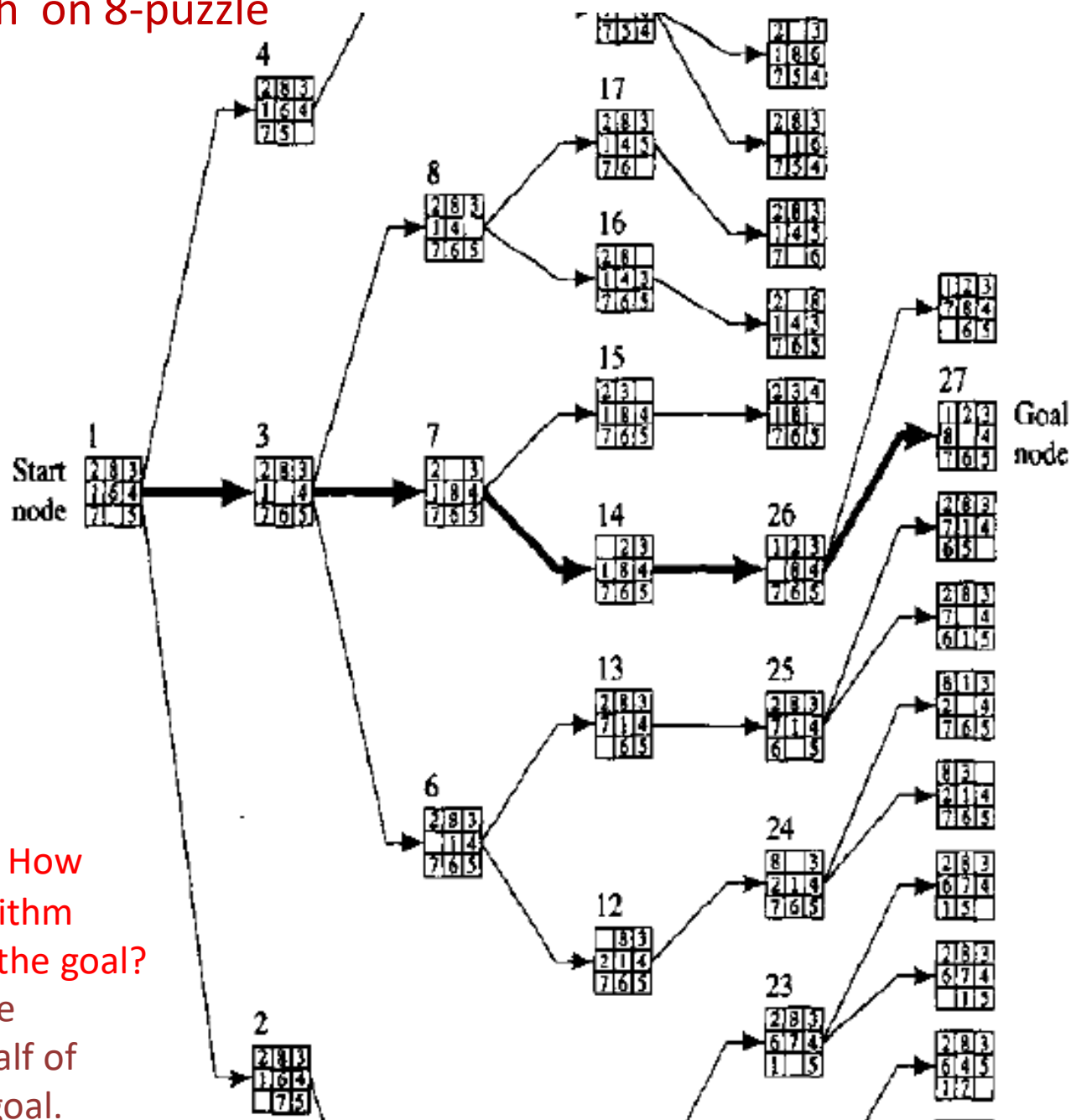
You can use Chinese words here if that's easier.

Healthy Perspective of AI

- The purpose of AI is to improve/innovate technology
- Many tasks, humans' way is good, provide inspiration.
- Some tasks humans don't do well:
 - Compare two pictures of the same person to check if they are some person
- The over-all drive is to develop new/effective technology to solve more tasks
 - Some used to be done by humans
 - Many new tasks, like new city planning, control, etc
- Other technology such as using big data, data driven methods are not necessarily AI.

Homework Intro2: Can you think of a task humans don't do well.

Breadth-First Search on 8-puzzle



Homework-BFS1: How many states the algorithm have searched to get the goal?

Assume on level 5, the algorithm searched half of the states to get the goal.

DFS: Recursive calls to DFS-VISIT()

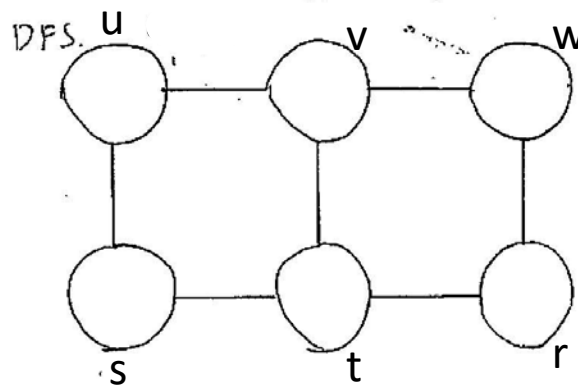
Homework:DFS1

1. Draw the DFS search tree on this graph.

Question: Vertex 1 has two adjacent vertices. Who will be visited first?

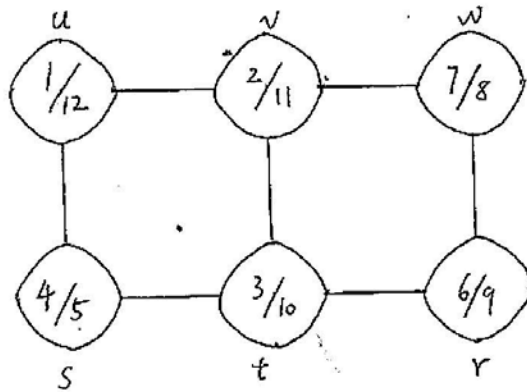
Answer: Depends on the order links in the adjacency list.

DFS: Recursive calls to DFS-VISIT()



Graph Representation
edgelist (undirected)

- (u v)
- (u s)
- (v t)
- (v w)
- (s t)
- (t r)
- (w r)



```

1. DFS-VISIT(u)
2.   DFS-VISIT(v)
3.     DFS-VISIT(t)
4.       DFS-VISIT(s)
5.         EXIT(s)
6.       DFS-VISIT(r)
7.         DFS-VISIT(w)
8.           EXIT(w)
9.         EXIT(r)
10.      EXIT(t)
11.    EXIT(v)
12.  EXIT(u)
    
```

Backtrack from s

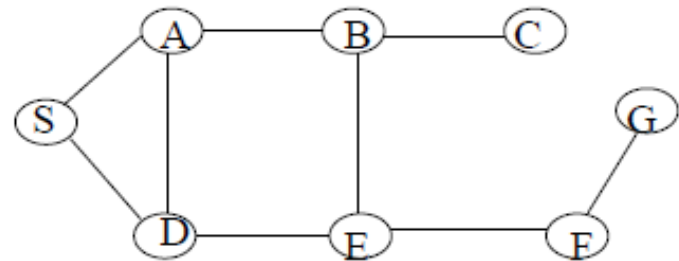
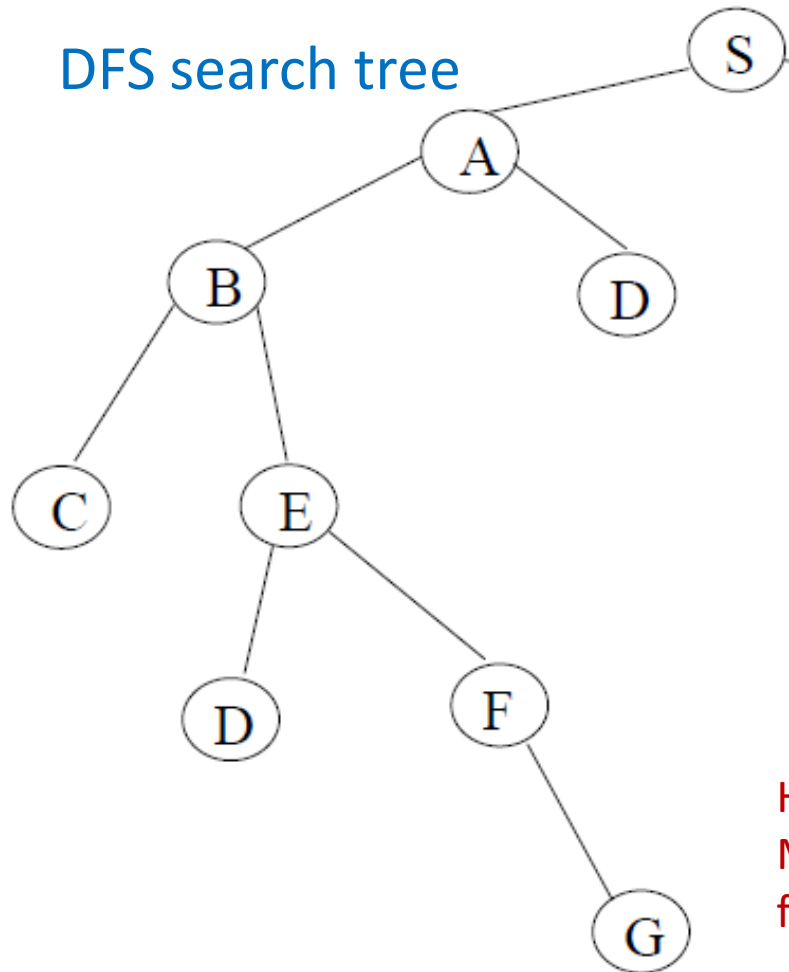
Backtrack from w

Backtrack from r

Backtrack from t

Depth-First Search (DFS)

DFS search tree



Homework:DFS2

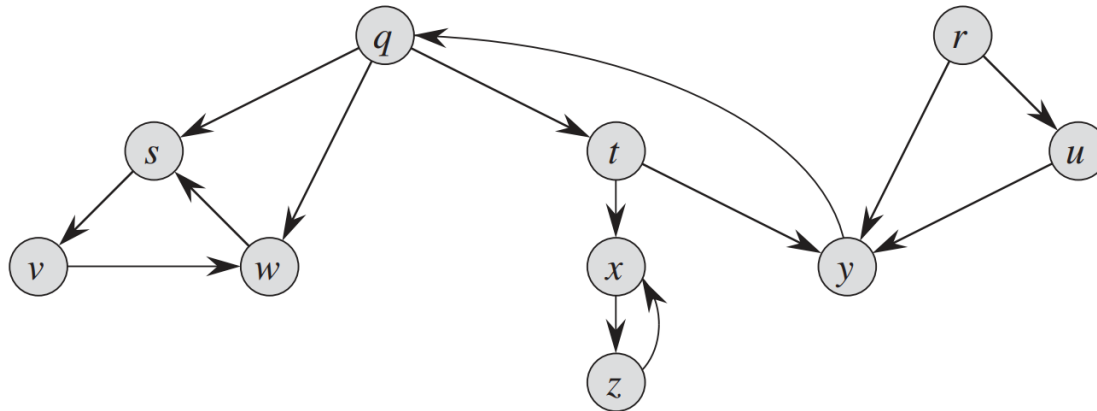
Mark the entering and leave time for every nodes.

Homework: DFS3

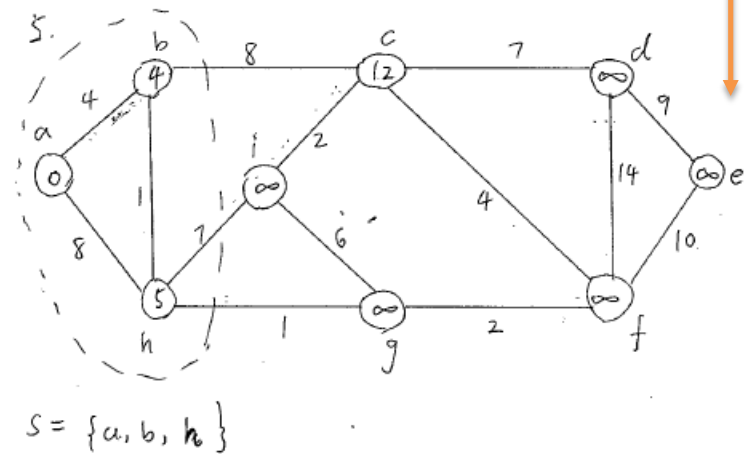
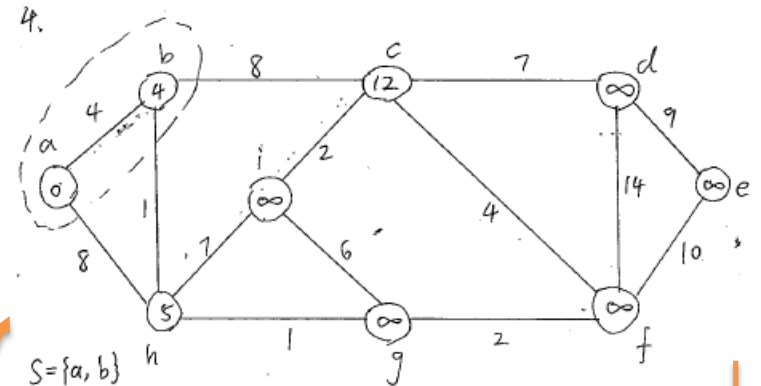
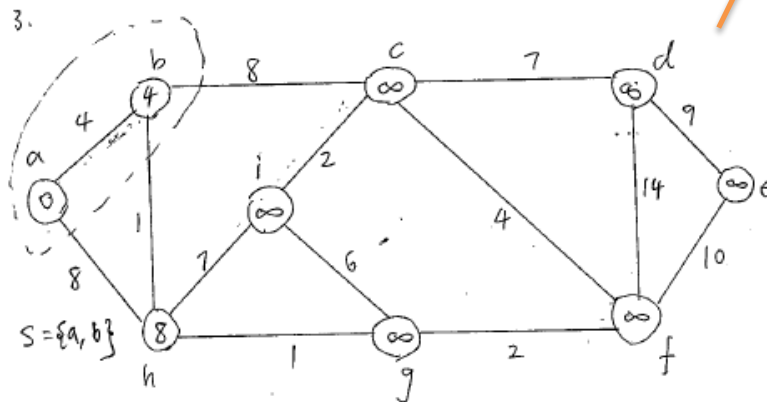
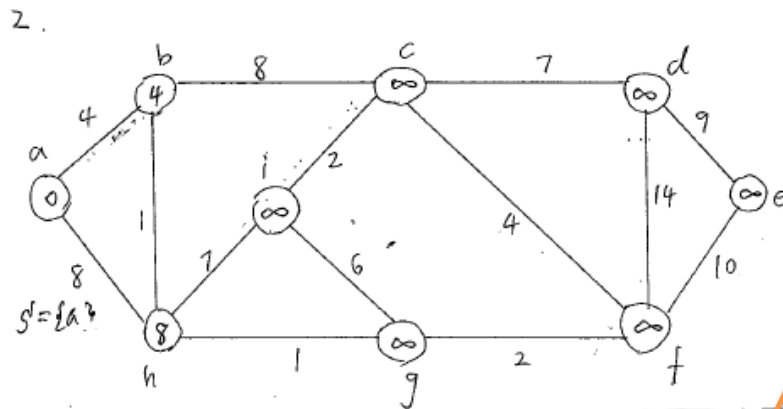
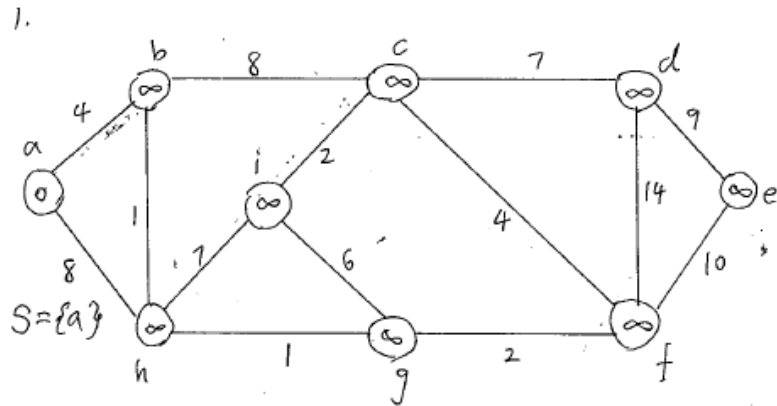
Show how DFS visit all nodes in the following graph.

Assume: In DFS(), the nodes are visited alphabetically.

Assume: The adjacency nodes of each nodes are listed alphabetically.

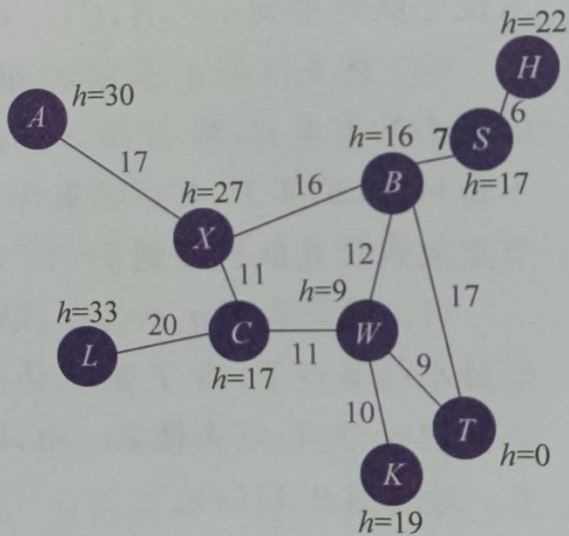


Another Example of Dijkstra Algorithm



HOMEWORK: Dijkstra1.
Complete the algorithm on this graph.

Homework A star 1: Running A* on the graph. A is start, T is goal. Your task: add $g(n)$ and $f(n)$ computation in each iteration of the while loop. See A* algorithm details.



初始化时, 优先队列为空 $q = \{\}$, 访问表为空 $v = \{\}$ 。算法执行过程如下:

(1) 将初始节点 A 放入优先队列得到 $q = \{A\}$ 。

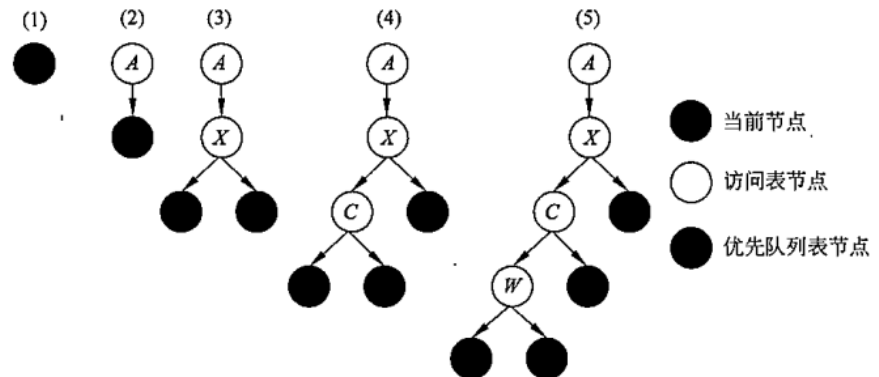
(2) 优先队列 q 不为空, 继续执行。弹出优先队列中 f 值最小的节点 A, 将 A 放入访问表得到 $v = \{A\}$ 。A 不是目标, 扩展 A, 其后继集合为 $\{X\}$ 。将后继集合中的节点放入优先队列得到 $q = \{X\}$ 。对应图 2.12(1)。

(3) 优先队列 q 不为空, 继续执行。弹出优先队列中 f 值最小的节点 X, 将 X 放入访问表得到 $v = \{A, X\}$ 。X 不是目标, 扩展 X, 其后继集合为 $\{B, C\}$ 。将后继集合中的节点放入优先队列得到 $q = \{B, C\}$ 。对应图 2.12(2)。

(4) 优先队列 q 不为空, 继续执行。弹出优先队列中 f 值最小的节点 C, 将 C 放入访问表得到 $v = \{A, X, C\}$ 。C 不是目标, 扩展 C, 其后继集合为 $\{L, W\}$ 。将后继集合中的节点放入优先队列得到 $q = \{W, B, L\}$ 。对应图 2.12(3)。

(5) 优先队列 q 不为空, 继续执行。弹出优先队列中 f 值最小的节点 W, 将 W 放入访问表 $v = \{A, X, C, W\}$ 。W 不是目标, 扩展 W, 其后继集合为 $\{T, K\}$ 。将后继集合中的节点放入优先队列得到 $q = \{T, B, K, L\}$ 。对应图 2.12(4)。

(6) 优先队列 q 不为空, 继续执行。弹出优先队列中 f 值最小的节点 T, 将 T 放入访问表得到 $v = \{A, X, C, W, T\}$ 。T 为目标节点, 结束搜索。由 A 到 T 的路径为 $A \rightarrow X \rightarrow C \rightarrow W \rightarrow T$ 。对应图 2.12(5)。

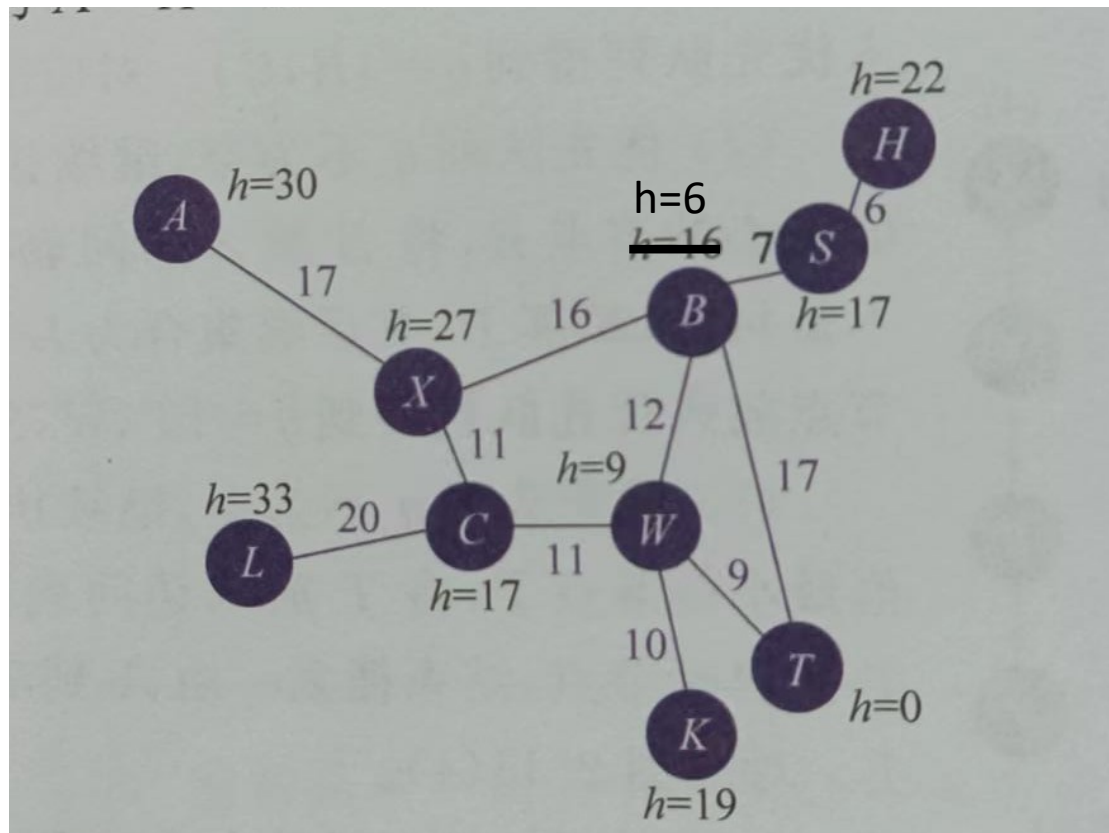


Homework A star 2: Running A* on the graph. A is start node, T is goal node. Here on node B, $h=16$ has been changed to $h=6$.

Can you get the same optimal path from A to T irrespective of whether $h=16$ or $h=6$ on node B? They are both admissible.

Give details as in **Homework A star 1**.

Is there some problem with the algorithm?



Homework: AStar3

Homework (optional):

Problem 3.7 on page 114 of the textbook. Solve two questions.

(a) How many states?

(b) Why shortest path goes along the edge of a polygon?

